

1st International Workshop on Data Analysis Patterns in Software Engineering (DAPSE 2013)

Christian Bird
Microsoft Research, USA
cbird@microsoft.com

Tim Menzies
West Virginia University, USA
tim@menzies.us

Thomas Zimmermann
Microsoft Research, USA
tzimmer@microsoft.com

Abstract—Data scientists in software engineering seek insight in data collected from software projects to improve software development. The demand for data scientists with domain knowledge in software development is growing rapidly and there is already a shortage of such data scientists. Data science is a skilled art with a steep learning curve. To shorten that learning curve, this workshop will collect best practices in form of *data analysis patterns*, that is, analyses of data that leads to meaningful conclusions and can be reused for comparable data. In the workshop we compiled a catalog of such patterns that will help experienced data scientists to better communicate about data analysis. The workshop was targeted at experienced data scientists and researchers and anyone interested in how to analyze data correctly and efficiently in a community accepted way.

Index Terms—data science, software engineering, data mining, predictive analytics, business intelligence, machine learning, big data, smart data, software analytics, software intelligence

I. INTRODUCTION

Software development generates an inordinate amount of data. For example, check-ins, work items, and test executions are recorded in software repositories such as CVS, Subversion, GIT, and Bugzilla. Telemetry data reflects how customers experience software, which includes application and feature usage and exposes reliability. Data scientists or researchers in software engineering seek insight in data collected from software projects in an effort to improve software development. Many companies such as Facebook, Google, Amazon, and Microsoft are actively working on becoming more data-driven in software development. The demand for data scientists with domain knowledge in software development is growing rapidly and there is already a shortage of such data scientists.

Data science is a skilled art with a steep learning curve. To shorten that learning curve, this workshop collected best practices in form of *data analysis patterns*, that is, analyses of data that leads to meaningful conclusions and can be reused for comparable data. In the workshop we compiled a catalog of such patterns that will help experienced data scientists to better communicate about data analysis and also newcomers get up to speed quickly and avoid common pitfalls.

The goals of the workshop were:

- Build a community interested in sharing data science expertise, for example by demonstrating data analysis, sharing of methodological decisions, setbacks encountered, and insights into their analysis techniques.

- Compile a catalog of reusable data analysis patterns that will be publicly available.
- Inform attendees regarding analysis techniques so that they are able to better understand when and how to use various data analysis techniques correctly and efficiently in a community accepted way.

The workshop was targeted to anyone who is interested in data analysis and wants to educate themselves in what analysis techniques are available and recommended.

For more information, please visit the workshop webpage:
<http://research.microsoft.com/dapse2013/>

II. DATA ANALYSIS PATTERNS

We solicited short papers (2-3 pages) describing one or more data analysis pattern. Authors should use the form that is most suited to describe the pattern. Where possible, we encourage authors to describe pattern in a similar way to the Gamma et al. design patterns book:

- Pattern name: a handle for the pattern
- Problem: when to apply the pattern
- Solution: how to apply the pattern
- Discussion: results and trade-offs of applying the pattern, common mistakes in applying the pattern that should be avoided, etc.
- Examples: briefly summarize and/or cite example applications of the pattern in literature, if possible, R snippets or Weka code to apply the pattern, etc.

For illustrative purposes, we provided an example pattern (see Figure 1) in short and simplified form in the call for papers; for the workshop we expected discussion of patterns to be more comprehensive.

We allowed two options for submitting a pattern:

- Archival Papers had to be submitted by the workshop paper deadline (February 7, 2013) and if accepted, they were published in the workshop proceedings and the ACM and IEEE Digital Libraries.
- Non-Archival Papers had a later deadline and were *not* published in the workshop proceedings and the ACM and IEEE Digital Libraries.

Pattern name: Contrast

Problem: Determine if there is a difference in one or more *properties* between *two populations*.

Solution:

1. Apply a hypothesis test (student t-test for parametric data, Mann Whitney test for non-parametric test) to check if the property is statistically different between populations.
2. Determine the magnitude of the difference, either through visualization (e.g., boxplot) or when appropriate through mean or median.

Discussion:

Either step without the other can be misleading. For large populations, tiny differences might be statistically significant. In contrast for small populations large differences might not be statistically significant.

Choosing the wrong hypothesis test is a common mistake.

Examples: For example, at ICSE 2009, Bird et al. used a Mann Whitney test to compare the defect proneness (=the property) between distributed and co-located binaries (=two populations). See Figure 5 in their paper for a sample visualization of the differences between the two populations.

Figure 1. Sample pattern provided in the call for papers.

Both archival and non-archival papers were reviewed by a program committee and accepted based on the clarity of the description and how broadly their proposed pattern might be applicable. Prior application of the pattern by the authors will not be a requirement and this workshop is more interested in the mechanics and choice of the data analysis than the impact of published results.

III. BACKGROUND OF THE ORGANIZERS

Christian Bird is a researcher at Microsoft Research (Redmond, United States). He received his PhD from UC Davis under advisor Prem Devanbu. His main research interest is empirical software engineering, predominantly examining collaboration and coordination in large software teams in both industrial and open source contexts. He has been involved in many data analysis studies, most recently examining code movement in SCMs. His studies have been published at venues including CACM, ICSE, FSE, MSR, and ICSM and he is the

recipient of an ACM Distinguished paper award. Christian has served on the program committees at FSE, MSR, ICSM, ICPC, and numerous workshops and has reviewed for CACM, TSE, TOSEM, EMSE, JSS, ICSE, CSCW, and others. He organized the Software Analytics track at the 2012 Microsoft Research Faculty Summit, is currently helping organize the Workshop on Replications in Software Engineering Research (RESER), and is co-chair of the Industry track at ICPC 2013

Tim Menzies is a Full Professor at WVU. He has 200+ refereed publications. His experience in data analysis is extensive. Since, 2001 he has been one of the leading proponents of applying data mining to software engineering data. He is the inventor of two new data mining algorithms (TAR3 and KEYS2) and is one of the co-organizers of the PROMISE conference on data mining SE data. He has organized workshops for ICSE 1999; ICSE 2005; ICSE 2007, ICSE 2012 and co-located conferences for ICSE 2008; ICSE 2009. He was the PC co-chair for ASE 12. A member of the editorial boards of IEEE TSE, ESE, JLV, ASE journal, he has edited special issues for ESE journal (five times); RE journal; and IEEE Intelligent Systems.

Thomas Zimmermann is a researcher in the Research in Software Engineering (RiSE) group at Microsoft Research, adjunct assistant professor at the University of Calgary, and affiliate faculty at University of Washington. His research interests include empirical software engineering, mining software repositories, development tools, social networking, and computer games. He is best known for his work on systematic mining of version archives and bug databases to conduct empirical studies and to build tools to support developers and managers. He received three ACM SIGSOFT Distinguished Paper Awards for his work published at the ICSE '07, FSE '08, and FSE'12 conferences. He has served on a variety of program committees, including ICSE, OOPSLA, ECOOP, ISSTA, and MSR. He was co-chair of the program committee for MSR '10 and '11. He co-organized the DEFECTS workshop (2008 and 2009) and the RSSE workshop (2008, 2010, and 2012). He received his PhD degree from Saarland University, Germany. For more information, visit <http://thomas-zimmermann.com/>